

The Effect of Surprisal on Reading Times in Information Seeking and Repeated Reading | CoNLL 2024

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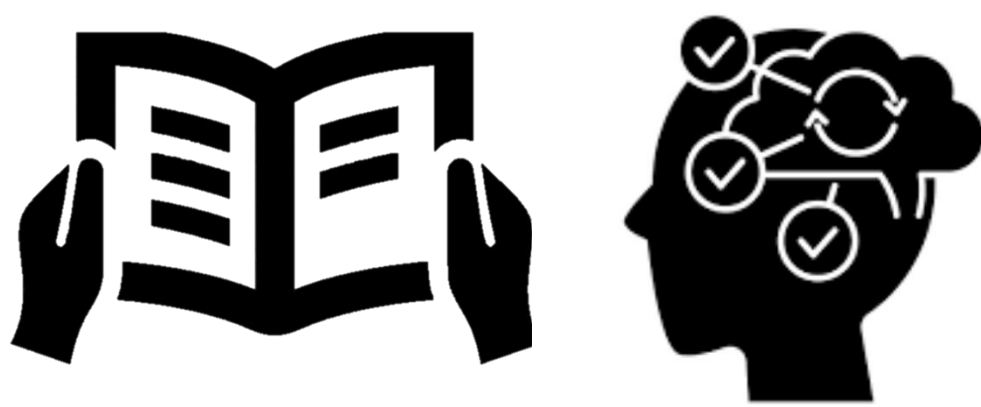
FACULTY OF DATA
AND DECISION SCIENCES
MAKING SENSE OF IT



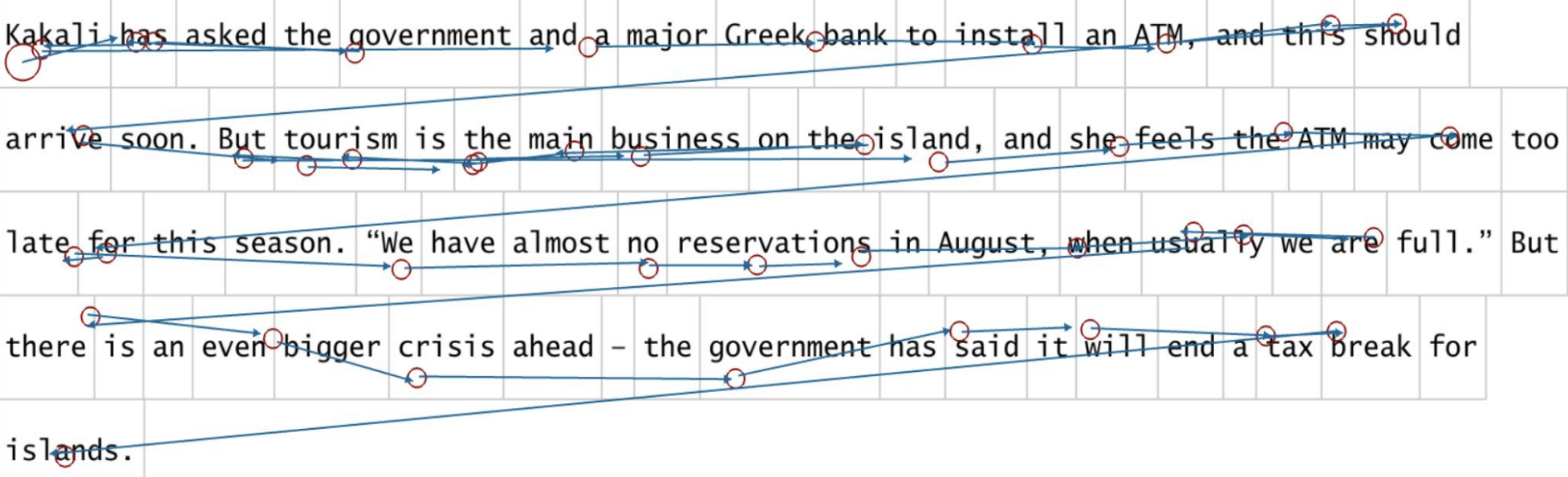
In-context memorization makes LLM surprisal useless for modeling repeated reading

Introduction

To what extent are language models useful for predicting the processing difficulty of humans?



Reading times as a proxy for processing difficulty



Surprisal Measure

$$\text{surprisal}(w_i) := -\log_2 p(w_i | w_{<i})$$

Context

Example: The dog chased the lion → High Surprisal



LLM
Surprisal

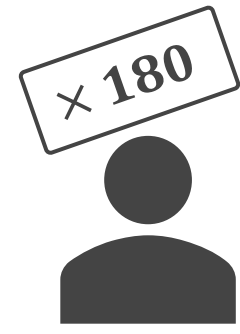


Human
Reading Times

This Study:

Using LLM surprisal estimates to predict reading times of native English speakers in a repeated reading setting

OneStop Eye Movements Dataset



Each Participant reads:



First Reading



Repeated Reading



Article 1



Article 2



Article 10



Article 11



Article 12

In total $\approx 550k$ Data Points

Word	Surprisal	Reading Time	...
lion	12	280	

Predict Reading Times (RT)

Generalized Additive Models (GAM)

Baseline $RT \sim te(freq, len) + te(freq_{prev}, len_{prev})$

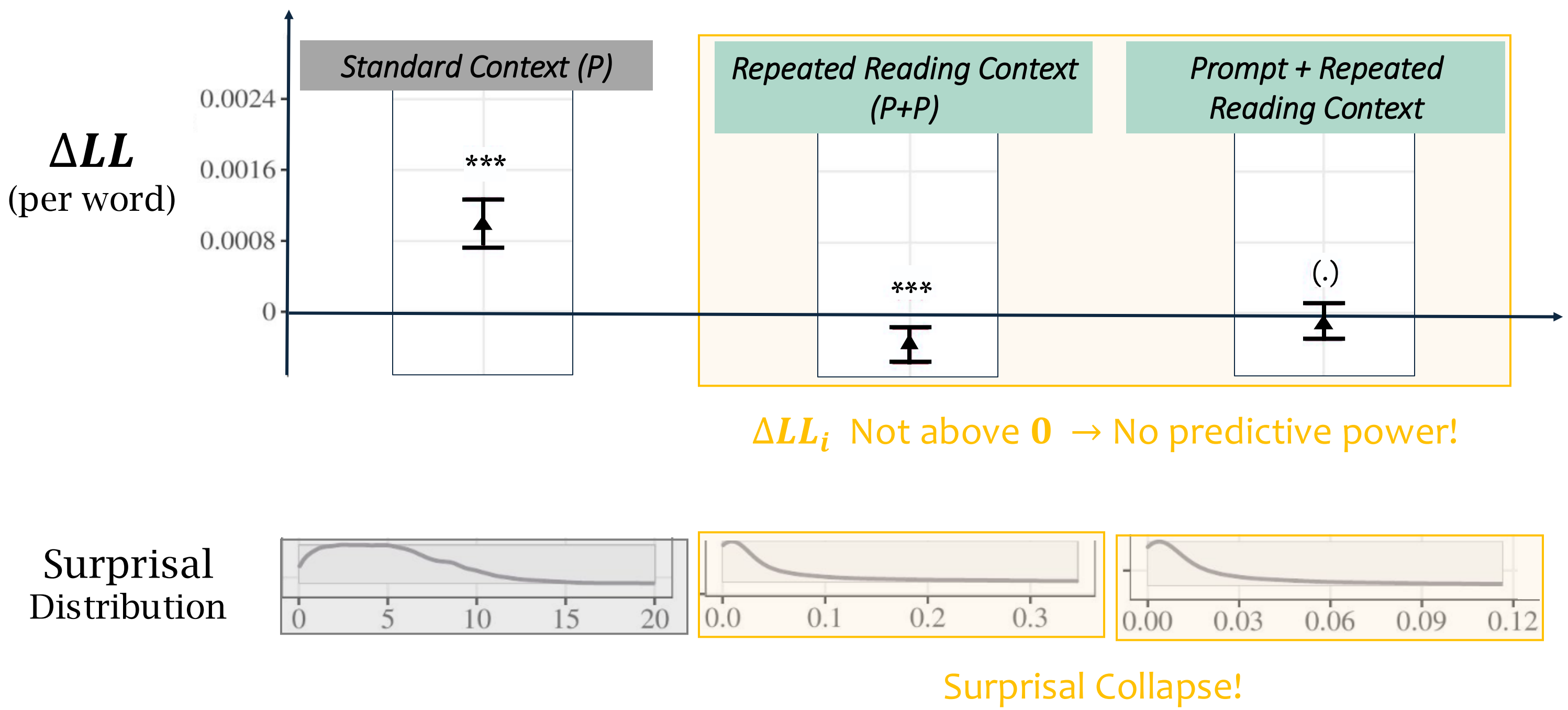
Target $RT \sim s(surp) + s(surp_{prev}) + te(freq, len) + te(freq_{prev}, len_{prev})$

Predictive Power for RT

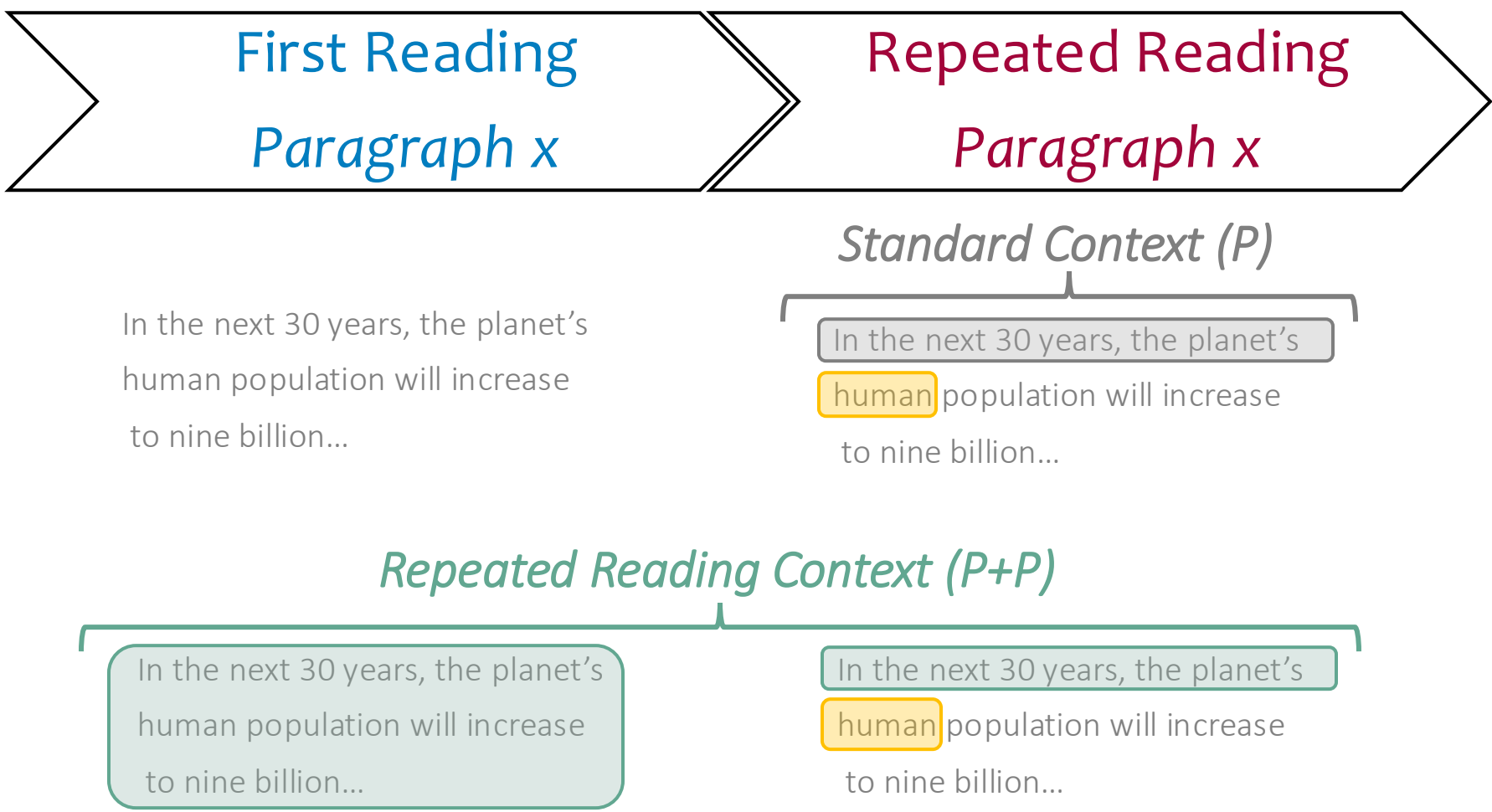
$L^M(RT_i | x) := \text{Likelihood of a RT value for a word with surprisal } x, \text{ under a model } M$

$$\Delta LL_i = \log L^{\text{target}}(RT_i | x^{\text{target}}) - \log L^{\text{baseline}}(RT_i | x^{\text{baseline}})$$

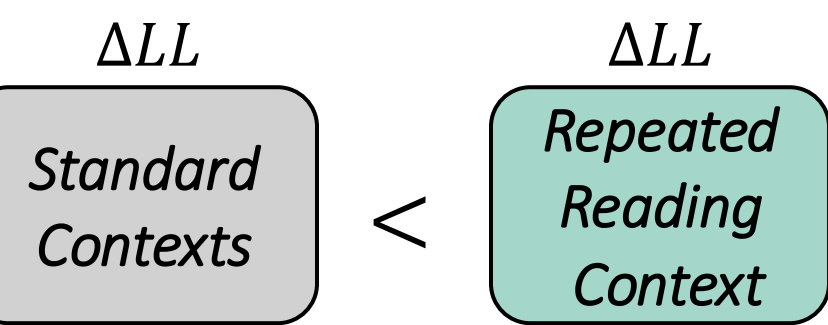
Results – Predictive Power



Repeated Reading Context

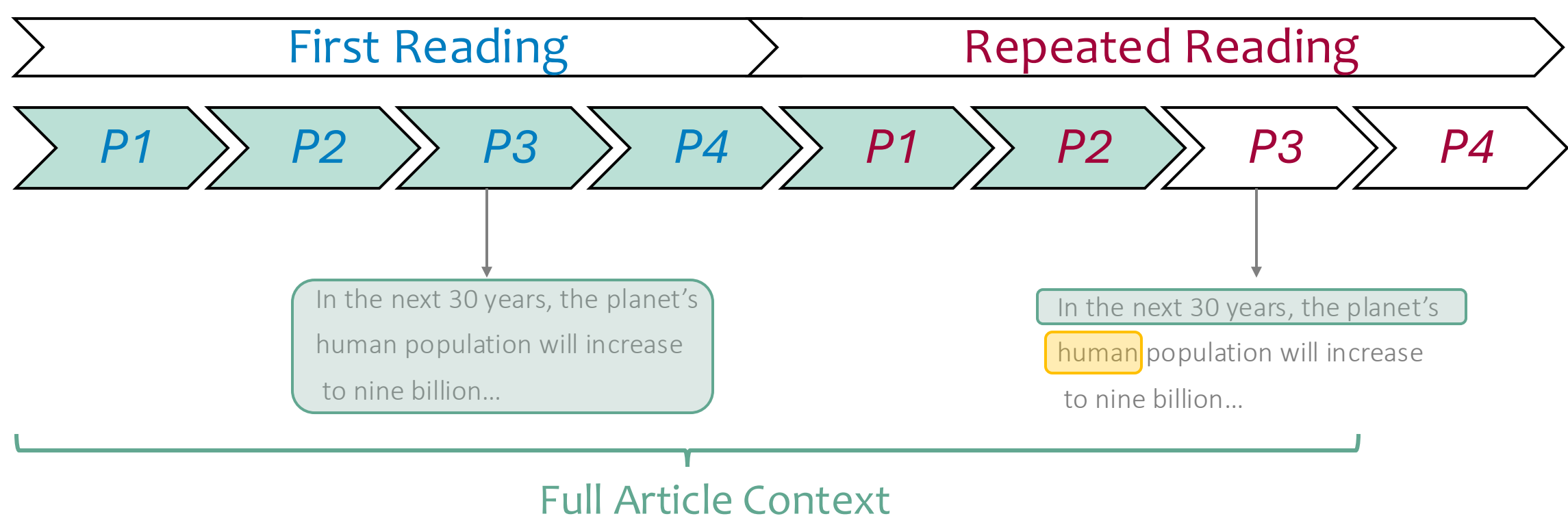


Hypothesis

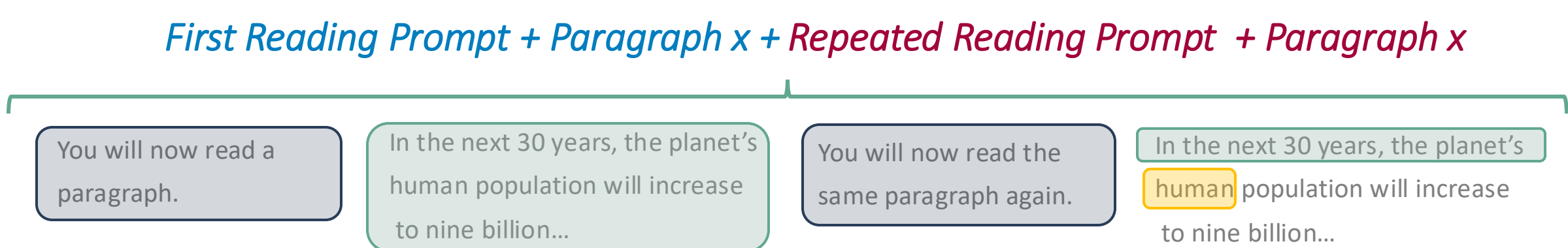


Higher predictive power for reading times when using *repeated reading context*

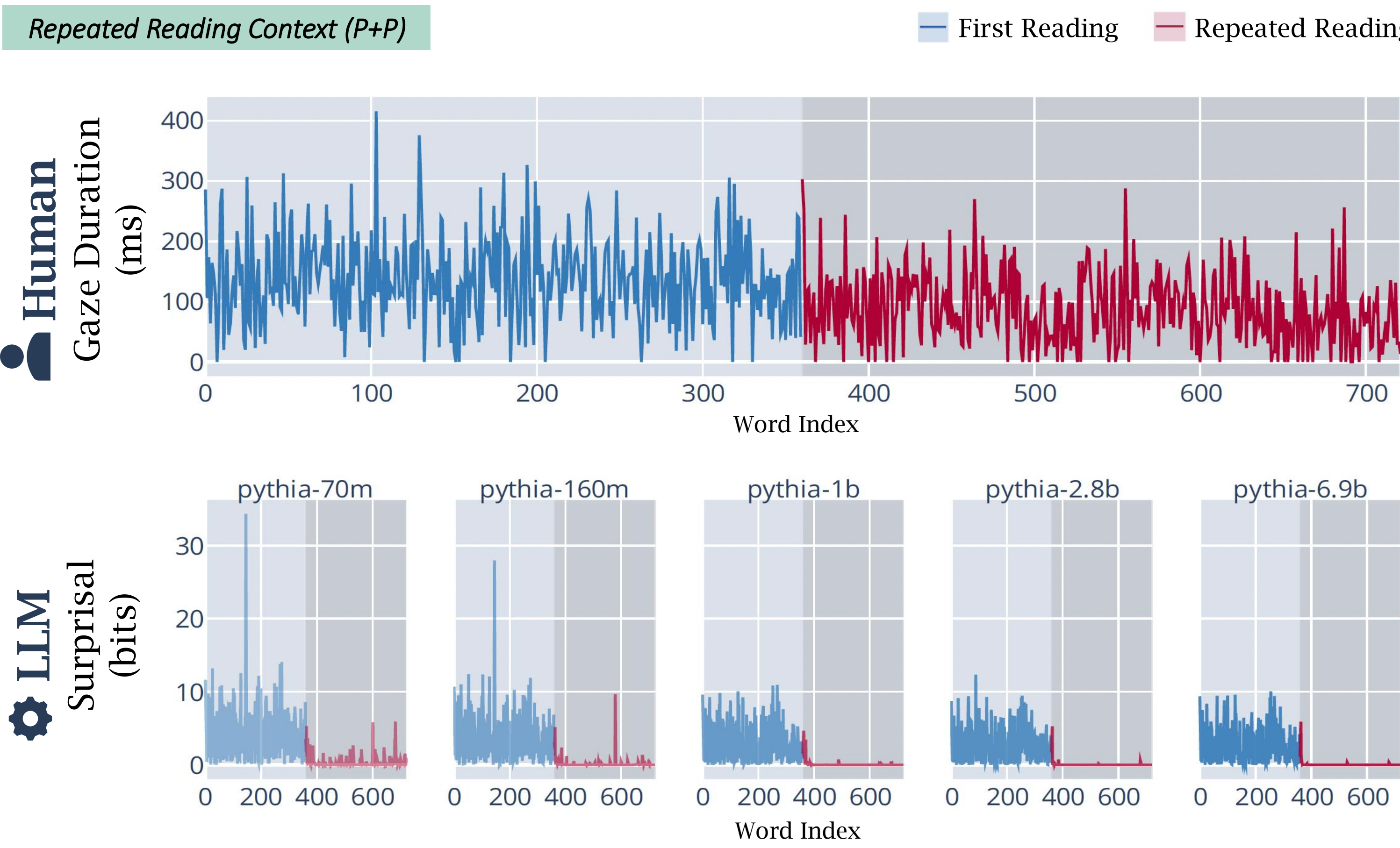
Include Intervening Material



Prompt + Repeated Reading Context



Results – Repeated Text Memorization



Conclusions

- The examined language models are **misaligned** with human reading patterns in **repeated reading**.
- Further research is needed for characterizing the factors that drive **in-context memorization**, and for developing more **cognitively plausible** memory mechanisms in LLMs.